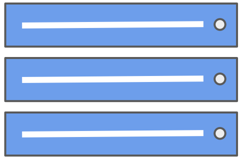
DeepLearning.AI

Model Serving Architecture

Model Serving: Patterns and Infrastructure

ML Infrastructure

On Prem



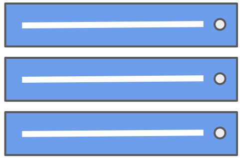
- Train and deploy on your own hardware infrastructure
- Manually procure hardware GPUs, CPUs etc
- Profitable for large companies running ML projects for longer time

On Cloud

- Train and deploy on cloud choosing from several service providers
    - Amazon Web Services, Google Cloud Platform, Microsoft Azure, etc

Model Serving

On Prem



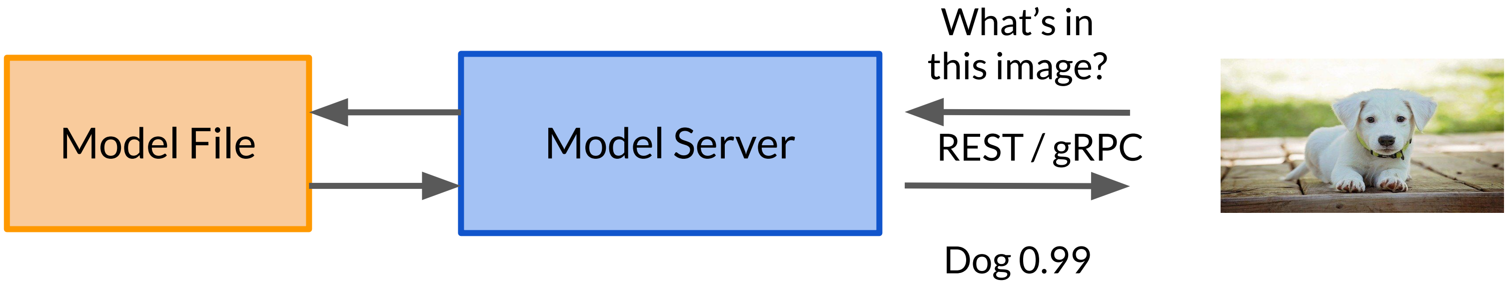
- Can use open source, pre-built servers
    - TF-Serving, KF-Serving, NVidia and more...

On Cloud

- Create VMs and use open source pre-built servers
- Use the provided ML workflow

Model Servers

- Simplify the task of deploying machine learning models at scale.
- Can handle scaling, performance, some model lifecycle management etc.,



Model Servers

TensorFlow

TensorFlow Serving

PyTorch

TorchServe

Kubeflow

KF Serving

NVIDIA

TRITON INFERENCE SERVER

TensorFlow Serving

TensorFlow

Supports many servables

TF Models	Non TF Models	Word Embeddings	Vocabularies	Feature Transformations

Out of the box integration with TensorFlow Models

Batch and Real-time Inference

Multi-Model Serving

Exposes gRPC and REST endpoints

Torch Serve

- Model serving framework for PyTorch models.
- Initiative from AWS and Facebook

PyTorch

TorchServe

Batch and Real-time Inference

Supports REST Endpoints

Default handlers for Image Classification, Object Detection, Image Segmentation, Text Classification

Multi-Model Serving

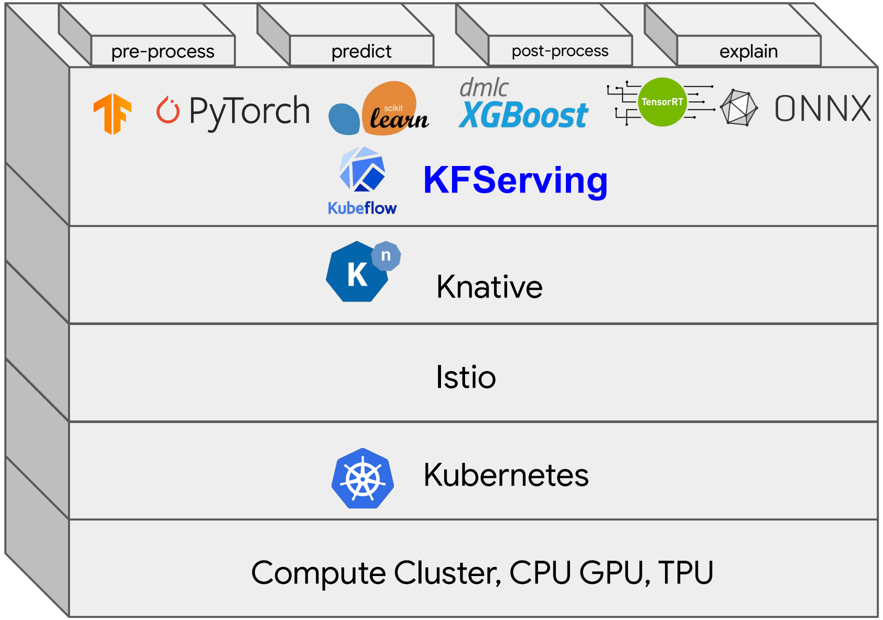
Monitor Detail Logs and Customized Metrics

A/B Testing

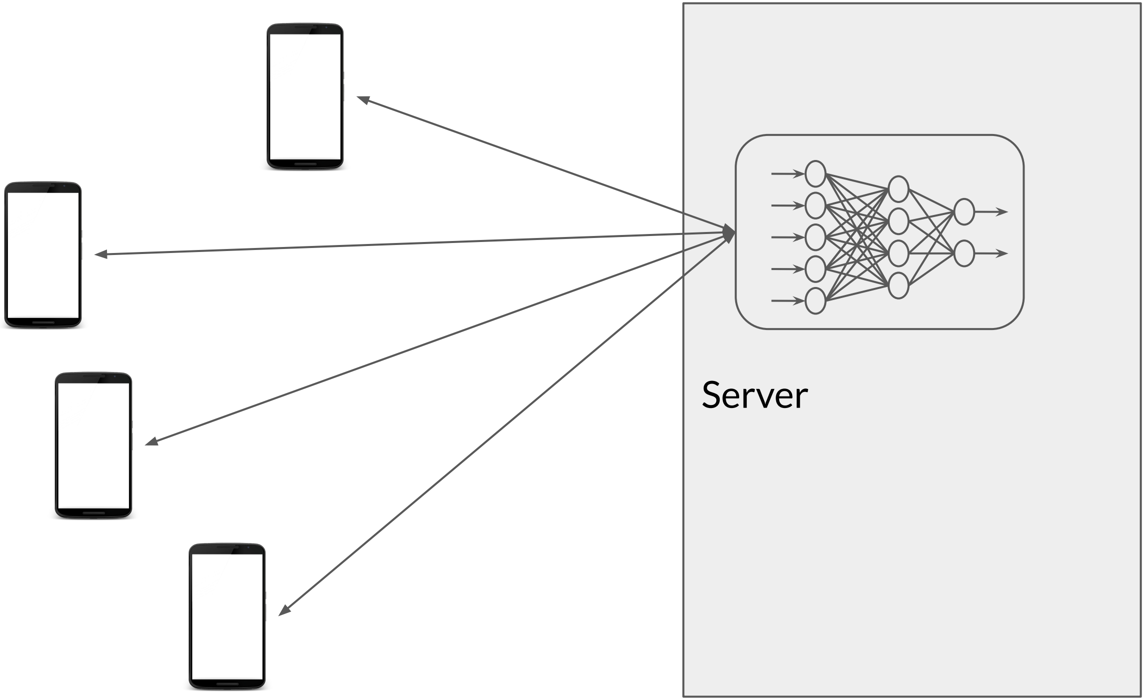
KFServing

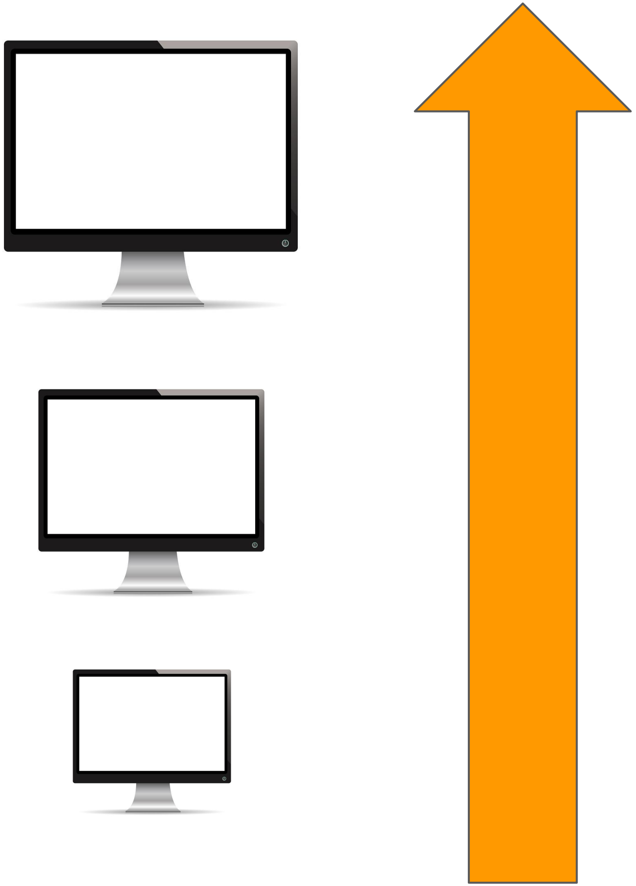
- Enables serverless inferencing on Kubernetes.
- Provides high abstraction interfaces for common ML frameworks like TensorFlow, PyTorch, scikit-learn etc.

KF Serving

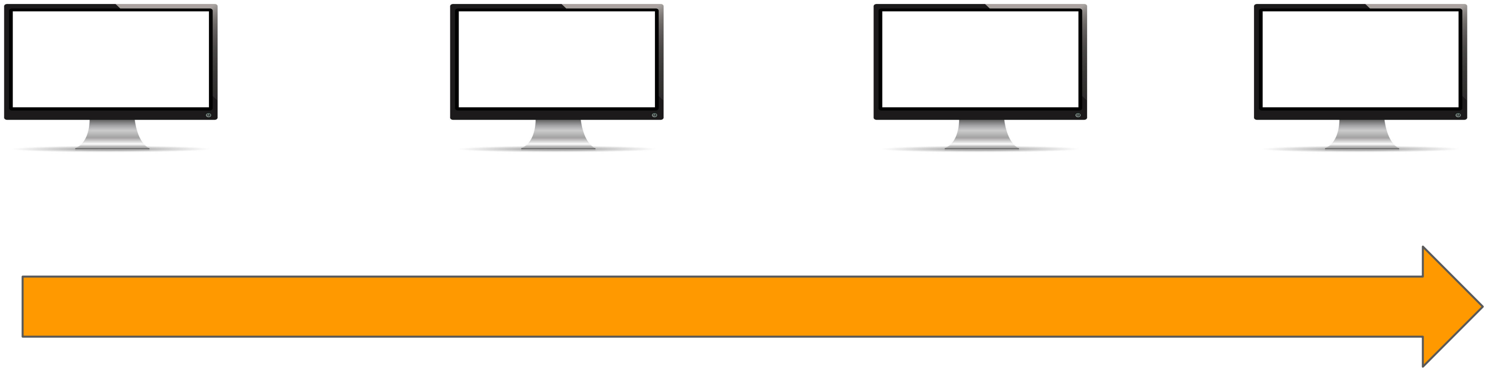


Why is Scaling Important?





- Increased Power
- Upgrading
- More RAM
- Faster Storage
- Adding or upgrading GPU/TPU



+ More CPUs/GPUs instead of bigger ones
+ Scale up as needed
+ Scale back down to minimums

Why Horizontal Over Vertical Scaling

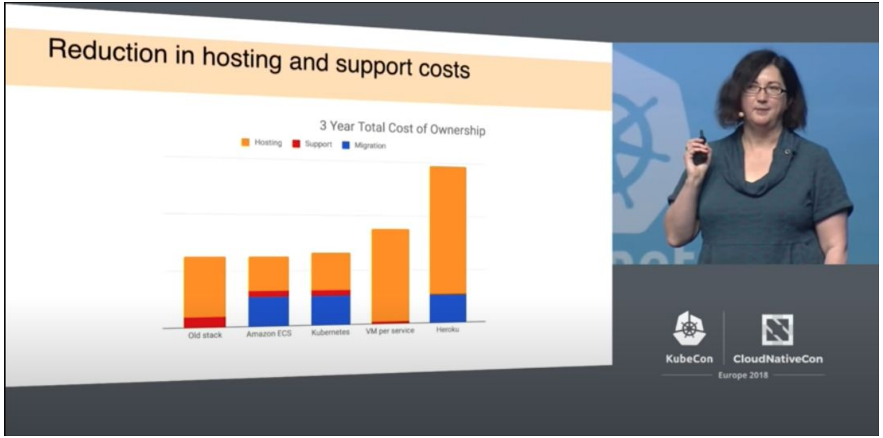
Benefit of elasticity	• Shrink or grow no of nodes based on load, throughput, latency requirements.
Application never goes offline	• No need for taking existing servers offline for scaling
No limit on hardware capacity	• Add more nodes any time at increased cost

Popular Container Orchestration Tools

Kubernetes

Docker Swarm

Kubernetes



Reduction in hosting and support costs
3 Year Total Cost of Ownership
Hosting
Support
Migration
Old stack
Amazon ECS
Kubernetes
VM per service
Heroku
KubeCon
CloudNativeCon
Europe 2018

ML Workflows on Kubernetes - KubeFlow

- Dedicated to making deployments of machine learning (ML) workflows on Kubernetes simple, portable and scalable.
- Anywhere you are running Kubernetes, you should be able to run Kubeflow.
- Can be run on premise or on Kubernetes engine on cloud offerings AWS, GCP, Azure etc.,